

Sample Examination
Production & Logistics
Winter Semester 2022/2023 – Heilbronn Campus

February xx, 2023, Moodle Examination

Round numbers to the second decimal (e.g., 0.455 to 0.46 or 0.454 to 0.45) unless otherwise stated.

Please note that the language in this exam is not indicative of answers, and (potential) grammar errors do not render an otherwise correct true statement into a false one.

Available time: 120 minutes.

Module: Production

Exercise 1: Multiple Choice Questions (12 Points)

In this exercise, each question answered correctly yields 1 point. Only one of the answers is correct, and there are no negative points for wrongly answered questions.

- 1.1** Because of an agreement with your supplier, your per-order setup cost has decreased by 19%. What effect it will have on your optimal order quantity, considering all other parameters remain same?
- a) Optimal order quantity increases by 10%.
 - b) Optimal order quantity decreases by 10%.
 - c) Optimal order quantity increases by 44%.
 - d) Optimal order quantity decreases by 44%.
- 1.2** You are using Economic Order Quantity model and need to decide on your order frequency in a finite planning horizon. As per your calculations, the optimum number of orders that must be placed in the finite period is a non-integer value. How must you decide on the number of orders that must be placed?
- a) Round up to the next integer value
 - b) Round down to the next integer value
 - c) Round to the integer value that minimizes the total cost
 - d) Data insufficient to decide.
- 1.3** Which of the following is a disadvantage of the Batch-and-Queue Production setup?
- a) Only a few products can be produced in a batch.
 - b) A large number of setups are required.
 - c) Less varieties of products can be offered.
 - d) Employees become jack of all trades, and master of none.
- 1.4** Which of the following is **TRUE** regarding Normal and Poisson distributions?
- a) Both distributions are continuous.
 - b) Both distributions are discrete.
 - c) The Normal distribution is discrete while the Poisson distribution is continuous.
 - d) The Normal distribution is continuous while the Poisson distribution is discrete.
- 1.5** Little's law gives the relationship between
- a) the average time to complete a process and the available capacity.
 - b) the average lead time for a process and the available buffer stock.
 - c) the average flow rate for a process and the average time.
 - d) the capacity utilization and the service level.

- 1.6** Which of the following forecasting techniques always leads to the worst forecasts?
- a) Judgmental forecasts
 - b) Time series forecasts
 - c) Associative forecasts
 - d) None of the above
- 1.7** Which of the following statement is **TRUE** when a company always orders an amount which equals exactly to Economic Order Quantity?
- a) The company incurs higher annual inventory holding costs compared to annual ordering costs.
 - b) The company incurs lower annual inventory holding costs compared to annual ordering costs.
 - c) The company produces exactly to the demand and works in accordance to lean philosophy.
 - d) The average annual inventory holding costs and annual ordering costs become same.
- 1.8** For smoothing constant $\alpha = 1$, exponential smoothing forecast becomes the same as
- a) Moving average forecast
 - b) Linear Regression forecast.
 - c) Naïve with trend forecast.
 - d) Naïve forecast.
- 1.9** Consider that number of new startups per week in Germany follows a Poisson distribution with a mean $= \mu$. The standard deviation is given by
- a) $1/\mu$
 - b) μ^2
 - c) $\sqrt{\mu}$
 - d) $1 - e^{-\mu}$
- 1.10** Which of the following is **NOT TRUE** for any distribution of a random variable?
- a) The probability density function can be zero.
 - b) The cumulative density function is (weakly) monotonically increasing.
 - c) The cumulative density function can be infinity.
 - d) The probability density function is below 1.
- 1.11** Which of the following is a part of the lean transformation?
- a) Allowing backorders in the inventory models
 - b) Using fool-proofing to reduce errors in a process execution (Poka Yoke)
 - c) Produce in larger batches to reduce the setup time

d) Building large inventory to fulfill even sudden increases in demand

1.12 Which of the following relates to Kaizen?

- a) Saving costs
- b) Produce continuously
- c) Continuous improvement
- d) Radical change

Exercise 2: Forecasting (12 points)

2.1 Consider the following time series to answer the following questions.

Period	Actual demand	Exponential Smoothing	Moving Average	Linear Regression
Sunday	336			
Monday	494			
Tuesday	360			
Wednesday	748			
Thursday	858			
Friday	596			
Saturday	1122			

- i. Assume $\alpha = 0.63$ for Exponential Smoothing with an initial forecast the same as the demand for Sunday. Calculate the exponential forecast for Thursday. (1)
- ii. Use actual demand data from the past two days to calculate the moving average forecast for Saturday. (1)
- iii. Find the slope and demand intercept of the straight line that best fits the demand curve. (2)

Slope:

Demand intercept:
- iv. Which of the following forecasting technique must be selected for the given dataset if your decision criteria are MSE? (2)
 - a) Moving average
 - b) Exponential smoothing
 - c) Linear regression
 - d) Any of three methods can be chosen.
- v. Find the optimal smoothing constant for the given dataset considering MAD as your decision criteria for selection. (2)

2.2 Why is forecasting required for a company before placing order to its suppliers? Provide 3 arguments. (3)

2.3 The sum of squared errors for five periods of a forecast is 125, and this forecast has no bias. What should be the control limits considering the confidence interval of 97.725%? (1)

- a) Upper control limit: 10 & lower control limit: -10
- b) Upper control limit: 62.5 & lower control limit: -62.5
- c) Upper control limit: 22.36 & lower control limit: -22.36
- d) Upper control limit: 9.77 & lower control limit: -9.77

Exercise 3: Capacity Management and Process Analysis (12 points)

3.1 Consider the process of registration of a new inhabitant at a Townhall. The visitors arriving the Townhall have to perform Check-in, which also involves showing Covid vaccination certificate. This activity takes 3 minutes per customer. The visitor is provided with a token and have to wait for 8 minutes in a queue before their turn for document checks. The document checks takes another 5 minutes. After the document check, the visitor have to wait for three more minutes before making the payment. The payment process takes 3 minutes. After the payment is made, the City Registration Certificate is prepared, which takes 10 minutes. Once the Certificate is ready, the visitors can show their token, and can collect their certificate. This process takes 2 minutes. The visitor then leaves the Townhall.

There are four groups of workers employed at the Townhall. There are two Group A employees, who performs Check-in and Check-out activities (giving token and sending out certificates). Two Group B employees are responsible for document checks. One Group C employee carry out payments, and three Group D employees prepare certificates. Group D employees, however, can do all the other jobs if required.

- i. What is the capacity of the Townhall? (2)
- ii. How many visitors are in a steady state simultaneously in Townhall? (2)
- iii. 14 individuals visit the Townhall every-hour to get the City Registration Certificate. Calculate the capacity Utilization of the Townhall. (2)
- iv. The Townhall employs one more Group D employee. Make optimum allocation of all Group D employees for different processes. (5)

Worker	Count	Check-in	Document Check	Payment	Certificate Preparation	Pickup
Group D	4					

- v. What is the new capacity of the Townhall? (1)

Exercise 4: Economic Order Quantity (12 points)

4.1 A distributor of sports equipment sells footballs to small retailers. It costs the distributor 98€ each time they want to place a new order from the manufacturer. Every six months, the distributor analyzes past demand and makes a forecast for the future. The distributor faces a stable demand of 5000 footballs every six months. The annual holding cost is €0.50/football.

- i. The distributor sources from Pakistan, which comes in a box of 5 balls, costing 15 €. These boxes of football are delivered immediately. How many boxes of football should be ordered by the distributor during each order cycle? (3)

Assume that the demand remains the same, and the distributor orders a quantity corresponding to the EOQ every time.

- ii. Calculate how many times the distributor should order if the selling season for the footballs is only 9 months at the beginning of the year; and the distributor does not want to keep a stock of un-sold footballs until the next selling season. Additionally, calculate how many boxes of footballs should be ordered by the distributor in this case. (4)

How many times the distributor should order? :

how many boxes of footballs should be ordered? :

The distributor is considering changing its supplier. The new supplier can supply the footballs in a box of 2 costing 6 €/box. The per-order setup costs, and the holding costs remains the same. However, the new supplier only delivers after 2.4 weeks of placing the order. (*Assume a month has 4 weeks*)

- iii. How many footballs must be in the inventory when an order is placed to the new supplier? (2)
- iv. What is the economic order quantity (of boxes) when ordering from the new supplier considering the same selling season? (3)

Exercise 5: Kristen's Cookie Company Case (6 points)

5.1 Consider Kristen's Cookie Company case:

- i. How can Kristen's Cookie Company improve its capacity? (1)
 - a) By buying a large fan for cooling cookies effectively.
 - b) By offering discounts for ordering cookies in large quantities.
 - c) By using an electric baking oven with 3 trays.
 - d) By incorporating an automated order processing and payment system.

- ii. Which of the following issues arises when the company starts delivering cookies directly from the apartment? (1)
 - a) More time required for delivery
 - b) Lower availability of cookies
 - c) Increased complexity of processes and losing the advantage of fresh-out-of-the-oven cookies
 - d) Less variety of cookies

- iii. Which production system is Kristen's Cookie Company using for preparing the order? (1)
 - a) Make-to-order
 - b) Engineer-to-order
 - c) Make-to-stock
 - d) Assemble-to-order

- iv. Kristen Explain why "Automating the order arrival process and payments do not improve the company's capacity." Also, suggest a recommendation to improve the capacity of Kristen's Cookie Company. (3)

Exercise 6: OceanCove Restaurant Case (6 points)

6.1 Consider the OceanCove restaurant case:

- i. Because of the new Omicron variant, the Government in India has prohibited dine-in. Businesses are not permitted to open before 9 a.m. or after 8 p.m. due to the limitations. Take-aways at restaurants are still possible. Given that the restaurant was operating from 11 AM to 11 PM before the Coronavirus outbreak, what effect do these restrictions have on the capacity and customer order lead-time? (2)
- ii. OceanCove has an option to select amongst different food delivery services which charge from 5%-10% of food price from the restaurant. What is the added value if OceanCove decides to a food delivery service? (2)
- iii. State any two disadvantages from an operations perspective if OceanCove decides to opt for multiple food delivery services? (2)

Module: Logistics

Exercise 1: Multiple Choice Questions (12 Points)

In this exercise, each question answered correctly yields 1 point. Only one of the answers is correct, and there are no negative points for wrongly answered questions.

- 1.1** The decision to adopt a pull or push production system is
- a) Operational
 - b) Tactical
 - c) Strategic
 - d) Financial
- 1.2** Which of the following is **NOT** a cause of the bullwhip effect?
- a) Seasonal discounts
 - b) Everyday low pricing
 - c) Inflated orders
 - d) Longer lead time
- 1.3** Which of the following is the primary objective of the Knapsack problem?
- a) To maximize the value of the knapsack.
 - b) To minimize the cost of the knapsack.
 - c) To maximize the weight of the knapsack.
 - d) To maximize the number of items in the knapsack.
- 1.4** What influence does the bullwhip effect have on order quantities as you move upstream in a supply chain?
- a) Increase in order volatility from customers to suppliers.
 - b) Order quantities remain constant across the supply chain.
 - c) Order quantities are independent of the bullwhip effect.
 - d) Decrease in order volatility from customers to suppliers.
- 1.5** What is the main idea behind “Hotelling Competition”?
- a) To dominate the competitor
 - b) To cooperate with the competitor in order to achieve the system-wide profit maximization by equally emphasizing ecologic, social, and governance aspects in a world driven by increased CSR demands.
 - c) To find an equilibrium solution concerning competitors.
 - d) To find an equilibrium solution concerning upstream/downstream companies.
- 1.6** Which distribution network design is used at a supermarket?
- a) Distributor storage with last-mile delivery
 - b) Distributor storage with carrier delivery
 - c) Retailer storage with customer pick-up

d) Distributor storage with package delivery

1.7 Which of the following is **NOT TRUE** for the Travelling Salesman Problem?

- a) Each node must be visited exactly once.
- b) The TSP solution is a cycle covering each node.
- c) The TSP solution is always the same as the shortest path problem.
- d) The TSP solution will necessarily repeat a node in its path.

1.8 In the order-up-to-inventory model, how can the inventory level be measured at the end of a period?

- a) The inventory level at the end of a period is the same as the order up-to level
- b) The inventory level at the end of a period is the same as the safety stock maintained by the company
- c) The inventory level at the end of a period is the same as the demand over the $l+1$ periods (l is the lead time from the supplier)
- d) The inventory level at the end of a period is the same as the order-up-to level minus the demand over the $l+1$ periods (l is the lead time from the supplier).

1.9 What is the rationale behind rounding down the optimum order quantity for the countable products in the newsvendor and order-up-to inventory models?

- a) By rounding down, companies could reduce their ordering costs
- b) By rounding down, companies could reduce their inventory holding cost
- c) By rounding down, companies could increase their expected benefits over expected losses
- d) Rounding down has a mainly practical motivation. It is simply easier for inventory managers just to drop the digits.

1.10 Shortage gaming is one of the causes of bullwhip effect. Please choose from the list below the category to which shortage gaming belongs, according to the lecture slides.

- a) Information distortion
- b) Behavioral factors
- c) Individual incentives
- d) None of the above

1.11 What happens to the service level if excess inventories can be salvaged for a positive unit price (less than marginal purchase costs) in the newsvendor model?

- a) The service level remains the same
- b) The service level decreases
- c) The service level increases
- d) The service level cannot be calculated

1.12 Which of the following is related to decision-making on a tactical level?

- a) Network configuration

- b) Production planning
- c) Operations strategy
- d) Production system

Exercise 2: Inventory Management (12 points)

2.1 You are operating a warehouse where you store phones. You place an order on a weekly basis, and the phones are delivered three weeks after placing the order. The weekly demand for these phones follows a Poisson distribution with a mean of 80. If the phones are out of stock, they can be back-ordered with a back-ordering cost of 20% of the phone's value per weekly backorder.

- i. The annual holding costs are {24-30}% of the phone's value. Calculate the optimal in-stock probability (in %). (2)
- ii. Find the optimal order-up-to- level corresponding to optimal service level that you find in (i). (3)
- iii. If (ii) is selected as the order-up-to level, what is the realized service level? (1)
- iv. How would the optimal service level change when the demand distribution is changed? (2)
- v. You want to change their inventory model from the order-up-to-inventory model to the newsvendor model. What changes should you expect while changing the inventory models? (1)
 - a) The excess items in stock can be used to fulfill the backorders.
 - b) There are no lost sales in the newsvendor model.
 - c) The newsvendor model focuses only on a single period.
 - d) Newsvendor model takes into consideration demands from past and upcoming periods.

2.2 Consider a product with newsvendor demand. The demand is uniform and continuously distributed between 250 and 450 units.

- i. The company incurs a cost of €3/unit and gains a profit margin of €{4-8}/unit. Find the optimum service level? (1)
- ii. Find the optimal number of units that are to be ordered? (2)

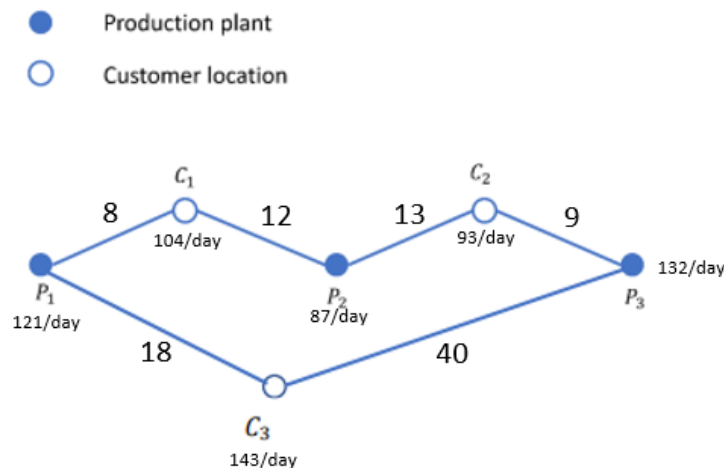
Exercise 3: Supply Chain Coordination (12 points)

3.1 Because of a trending social media video, the demand for Green Tea increased drastically. To meet the increasing demand and capitalize on the opportunity, retailers start placing increased orders to the suppliers, and the suppliers further to their suppliers. The manufacturer suddenly starts receiving large orders beyond its capacity.

- i. Explain how the sudden increase in consumer demand will affect the supply chain and who it will affect the most. (4)
- ii. What are the potential impacts of this phenomenon on supply chains? (mention 4 points) (4)
- iii. Which of the following mitigation strategies is not suitable to mitigate this phenomenon in this case? Explain. (4)
 - Pull system with make-to-order
 - Vendor-Managed Inventory
 - Electronic Data Interchange
 - Points of Sale Data

Exercise 4: Facility Location Problem (12 points)

- 4.1** A toy company serves three customers (C1, C2, and C3). Each plant can serve two different customers, as shown in the figure. The figure shows the distances between the locations and the capacity and demand of each production plant and customer location, respectively. The cost of producing each toy is €50, and the logistics costs are 4€/toy/km.



- i. Considering x_{ij} denotes the number of toys shipped from Plant i to Customer j , complete the following objective function that minimizes cost. (4)

$$\min \quad [\quad] x_{11} + [\quad] x_{13} + [\quad] x_{21} + [\quad] x_{22} + [\quad] x_{32} + [\quad] x_{33}$$

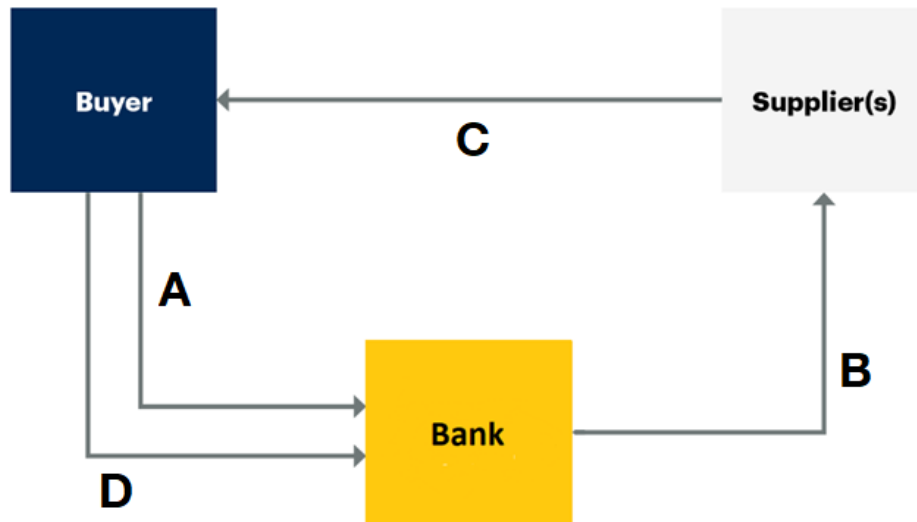
- ii. Solve this problem by applying the location covering problem to complete the following matrix: (6)

Amount to ship from plant i to customer j , x_{ij}		Customer		
		$j=1$	$j=2$	$j=3$
Plant	$i=1$		-	
	$i=2$			-
	$i=3$	-		

- iii. Calculate the optimal minimum cost (in €) the company will incur for fulfilling the demand of all the customer locations. (2)

Exercise 5: Supply Chain Finance (6 points)

5.1 Use the image below to answer the following questions:



- i. Explain different processes involved in reverse factoring. (4)
- ii. Find the correct sequence of reverse factoring. (2)

Exercise 6: HP Case (6 points)

6.1 Consider the HP case:

HP is producing its new large DeskJet printer at the Vancouver Plant. Owing to the new technological advantages as compared to other competitors, the demand for these printers is strong. The annual demand for these printers is 24,500 units, of which 30% comes from North America, 40% from Europe and the rest from Asia-Pacific region. The company's needs to decide if it should operate a European warehouse situated in the Netherlands, where the majority of customers are located, and can collect the printers. The cost of holding the printer inventory is \$400 per printer annually, excluding the cost of capital. The company incurs weighted average cost of capital of 22.5% of printer's net value, which is \$20,000. There are two modes of transportation:

- via Container ships – Duration: 11 weeks – Cost per container: €10,000 – Each container can hold 500 printers
- via Airfreight (direct to customer) – Duration: 2 weeks – Cost per item: €300

(Assume 52 weeks in a year)

- Find out which transportation model reduces the total cost for HP (including in-transit capital cost)? (2)
- Calculate how much HP can save with this selection (4)